THE HILBERT FUNCTION OF A FINITE GRADED MODULE IS A NUMERICAL POLYNOMIAL IN DEGREE ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Finiteness of the graded pieces over the degree-zero ring. Invoked in the statement itself at line 14115. Makes the classes of the graded pieces elements of the Grothendieck group.
- Numerical polynomials via the difference operator. Invoked at the end of the counted proof (line 14160). Converts the degree-by-degree difference computation into the polynomial conclusion.

Source reference. Original lemma 00K0: sources/stacks-algebra/pinned-original-source.tex, source lines 14090–14090. Navigation pointer for \ref{lemma-graded-module-fg} used inside the counted span (line 14115); the argument that consumes it is retained in the span.

Source reference. Original lemma 00JZ: sources/stacks-algebra/pinned-original-source.tex, source lines 14070–14070. Navigation pointer for \ref{lemma-numerical-polynomial} used inside the counted span (line 14160); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proposition 0.1. *Suppose that S is a Noetherian graded ring and M a finite graded S-module. Consider the function*

$$\mathbf{Z} \longrightarrow K'_0(S_0), \quad n \longmapsto [M_n]$$

see Lemma 00K0. If S_+ is generated by elements of degree 1, then this function is a numerical polynomial.

Proof. We prove this by induction on the minimal number of generators of S_1. If this number is 0, then $M_n = 0$ for all $n \gg 0$ and the result holds. To prove the induction step, let $x \in S_1$ be one of a minimal set of generators, such that the induction hypothesis applies to the graded ring $S/(x)$.

First we show the result holds if x is nilpotent on M. This we do by induction on the minimal integer r such that $x^r M = 0$. If $r = 1$, then M is a module over S/xS and the result holds (by the other induction hypothesis). If $r > 1$, then we can find a short exact sequence $0 \to M' \to M \to M'' \to 0$ such that the integers r', r'' are strictly smaller than r. Thus we know the result for M'' and M'. Hence we get the result for M because of the relation $[M_d] = [M'_d] + [M''_d]$ in $K'_0(S_0)$.

If x is not nilpotent on M, let $M' \subset M$ be the largest submodule on which x is nilpotent. Consider the exact sequence $0 \to M' \to M \to M/M' \to 0$ we see again it suffices to prove the result for M/M'. In other words we may assume that multiplication by x is injective.

Let $\overline{M} = M/xM$. Note that the map $x : M \to M$ is *not* a map of graded S-modules, since it does not map M_d into M_d. Namely, for each d we have the following short exact sequence

$$0 \to M_d \xrightarrow{x} M_{d+1} \to \overline{M}_{d+1} \to 0$$

This proves that $[M_{d+1}] - [M_d] = [\overline{M}_{d+1}]$. Hence we win by Lemma 00JZ. □